

AWS Practical Step-by-Step Guide

AWS setup and deployment steps — practical teacher-friendly format

IAM

How to Create an IAM User in AWS

Step 1: Open IAM

1. Search for “IAM” in the AWS Console and open IAM.

Step 2: Open Users

2. From the left menu, click “Users”.
3. Click “Create user”.

Step 3: Enter User Details

4. Enter the user name.
5. Click “Next”.

Step 4: Set Permissions

6. Select the required permission option.
7. **Attach the required policy:** You can assign specific permissions to the user based on their requirements. For example, if the user only needs access to EC2, attach an EC2-related policy. If the user only needs access to an S3 bucket, attach an S3-related policy. Similarly, you can attach policies for other AWS services. If the user requires full access to AWS services, select **AdministratorAccess** or add the user to the required group.
8. Click “Next”.

Step 5: Create User

9. Review the settings.
10. Click “Create user”.

2. How to Create IAM Access Keys -> this part of IAM User

Step 1: Open the IAM User

11. Open IAM → Users.
12. Select the required user.

Step 2: Open Security Credentials

13. Click the “Security credentials” tab.

Step 3: Create Access Key

14. Find "Access keys".
15. Click "Create access key".
16. Follow the prompts and create the key.
17. Save the Access Key ID and Secret Access Key securely.

How to Create an IAM Group

Step 1: Open Groups

18. Open IAM.
19. From the left menu, select "User groups".
20. Click "Create group".

Step 2: Enter Group Details

21. Enter the group name.
22. Select the required policies for the group.--> you can select permission according to scenario like (ec2,s3,full access etc)

Step 3: Create Group

23. Review the settings.
24. Click "Create group".
25. And you can add users to the group. Go to the **User groups** section and open the group you recently created. You will see an **Add users** button. Click it and select the users you want to add to the group based on the permissions they need.

How to Create an IAM Policy

Step 1: Open Policies

26. Open IAM.
27. Select "Policies" from the left menu.

Step 2: Create Policy

28. Click "Create policy".
29. Choose the required service and permissions.
30. Configure the resources and actions as required.

Step 3: Create Policy

31. Review the policy.
32. Enter a policy name.
33. Click "Create policy".

Step 1: Open IAM

34. Search for IAM in the AWS Console.
35. Open the IAM service.

Step 2: Open Policies

36. From the left-side menu, click Policies.
37. Here you can see the predefined policies already available in AWS.
38. These policies define the permissions that a user or group can have.
39. Or create a custom policy

Step 3: Create a Custom Policy

- Click **Create policy**.
- AWS allows you to create your own custom policy when the predefined policies do not meet your requirements.

Step 4: Select the Service

40. Select the AWS service for which you want to provide permissions.
41. For example, select EC2.
42. You can similarly select other services such as S3, RDS, VPC, etc.

Step 5: Select the Actions

43. After selecting the service, choose what actions the user or group is allowed to perform.
44. For example, for EC2, you can allow:
 - All EC2 actions
 - List actions
 - Read actions
 - Specific actions such as creating or deleting security groups
- You can make the permissions as specific as required.

Step 6: Select Resources

45. Specify which resources the policy should apply to.
46. You can select the required resources instead of giving access to everything.
47. Some AWS services may have dependencies on other services, so AWS may ask you to specify additional resources or permissions.

Step 7: Review the Policy

48. Click Next.
49. Review the permissions you selected.
50. AWS also shows the policy in JSON format.
51. You can review the JSON to see exactly which permissions have been configured.

Step 8: Create the Policy

52. Enter a Policy name.
53. Add a description if required.

54. Review all the permissions.
55. Click Create policy.

Step 9: Attach the Policy

56. After creating the policy, you can attach it to:
 - A specific IAM user
 - An IAM group
 - A role, when required
57. This policy determines what that user, group, or role is allowed to do in AWS.

How to Create an IAM Role

Step 1: Open Roles

58. Open IAM.
59. Select “Roles”.
60. Click “Create role”.

Step 2: Select Trusted Entity

61. Select the AWS service or trusted entity that will use the role.
62. Click “Next”.

Step 3: Add Permissions

63. Select the required policies.
64. Click “Next”.

Step 4: Create Role

65. Enter a role name.
66. Review the settings.
67. Click “Create role”.

EC2

How to Create an EC2 Instance

Step 1: Open EC2

68. Search for “EC2” in the AWS Console and open it.
69. Click on the instance
70. Click “Launch instance”.

Step 2: Enter Instance Details

71. Enter an instance name.
72. Select the required AMI (e.g linux)
73. Select the required instance type.--> now as a free account you can select t3.micro or t3 small etc but mostly t3.micro select for learning purpose

Step 3: Configure Key Pair

74. Select an existing key pair or create a new key pair.
75. If create the file then just click on the create new key pair button then will open one pop up then you give the name and select type like Rsa and then select format like .pem then just create and then automatically file download

Step 4: Configure Network (as it same i think not need to change this is security related)

76. Select the VPC and subnet.
77. Configure the security group.
78. Allow only the required traffic.

Step 5: Launch

79. Review the configuration.
80. Click "Launch instance".

1. How to Connect to an EC2 Instance

Step 1: Select Instance

81. Open EC2 → Instances.
82. Select the running instance.

Step 2: Connect

83. Click "Connect".
84. Go to In ssh client

Step 3: Connect Using SSH

85. Use the provided SSH command with your key pair.
86. Run the command from your terminal. (Note: agr command run karna sa kam ho jaa tu okay aur agr na hoe tu aso sa phal security group banoo i mean inbound banoo)

2. How to Create an EC2 Security Group

Note: You can create a Security Group before creating the EC2 instance and select it later during the instance creation process. In the **Network settings** section, click **Select existing security group** and choose the Security Group you already created, so there is no need to create a new one.

Alternatively, you can create a new Security Group directly while creating the EC2 instance. In simple words, **you can create a Security Group either before creating the EC2 instance or directly during the EC2 instance creation process.**

Step 1: Open Security Groups

87. Search EC2 and Open EC2.
88. Go to button side and see in left side **Network & Security**
89. There you see security groups
90. Select “Security Groups”.
91. Click “Create security group”.

Step 2: Configure Rules

92. Enter a security group name and description.
93. Select the VPC.
94. Add the required inbound rules.
95. Add the required outbound rules.

Step 3: Create

96. Review the rules.
97. Click “Create security group”.

How to Stop or Terminate an EC2 Instance

Step 1: Select Instance

98. Open EC2 → Instances.
99. Select the instance.

Step 2: Choose Action

100. Click “Instance state”.
101. Select “Stop instance” to temporarily stop it.
102. Select “Terminate instance” when the instance is no longer needed.

EC2 Networking Basis

12. How to Create a VPC

Step 1: Open VPC

103. Search for “VPC” in the AWS Console and open it.

Step 2: Create VPC

104. Click “Create VPC”.
105. Choose the required VPC configuration.
106. Enter the IPv4 CIDR block.
107. Enter a VPC name.

Step 3: Create

- 108. Review the configuration.
- 109. Click “Create VPC”.

13. How to Create a Subnet

Step 1: Open Subnets

- 110. Open VPC → Subnets.
- 111. Click “Create subnet”.

Step 2: Configure Subnet

- 112. Select the VPC.
- 113. Enter the subnet name.
- 114. Select the Availability Zone if required.
- 115. Enter the subnet IPv4 CIDR block.

Step 3: Create

- 116. Click “Create subnet”

S3 Bucket

1. How to Create an S3 Bucket

Step 1: Open S3

- 117. Search for “S3” and open it.
- 118. Click “Create bucket”.

Step 2: Configure Bucket

- 119. Enter a globally unique bucket name.
- 120. Select the AWS Region.
- 121. Configure the required public access settings.

Step 3: Create

- 122. Review the settings.
- 123. Click “Create bucket”.

2. How to Upload Files to S3

Step 1: Open Bucket

- 124. Open S3.
- 125. Select the required bucket.

Step 2: Upload

126. Click "Upload".
127. Add the required files or folders.
128. Click "Upload".

3. How to Enable S3 Versioning

Step 1: Open Bucket

129. Open S3 and select the bucket.

Step 2: Enable Versioning

130. Open the bucket's properties/settings.
131. Find "Bucket Versioning".
132. Enable versioning and save the change.

4. How to Enable S3 Object Lock

Step 1: Create/Select Bucket

133. Open S3 and select the bucket, or create a bucket with Object Lock enabled.

Step 2: Enable Object Lock

134. Open the Object Lock settings.
135. Enable Object Lock.
136. Configure the required retention settings.

5. How to Host a Static Website on S3

Step 1: Create Bucket

137. Create an S3 bucket with a globally unique name.

Step 2: Configure Website Hosting

138. Open the bucket.
139. Open "Properties".
140. Find "Static website hosting".
141. Enable it.
142. Set the index document, such as index.html.
143. Set the error document if required.

Step 3: Upload Website Files

144. Open "Objects".
145. Upload the website files.

Step 4: Configure Access

146. Configure the required bucket policy/public access settings for the website.

Step 5: Open Website

147. Use the S3 website endpoint shown in the bucket properties.

SNS Topic and Subscription

25. How to Create an SNS Topic and Subscription

Step 1: Open SNS

148. Search for “SNS” and open it.
149. Click “Topics”.

Step 2: Create Topic

150. Click “Create topic”.
151. Select the required topic type.
152. Enter a topic name.
153. And in other setting not need any changing
154. Create the topic.

Step 3: Create Subscription

155. Open the topic.
156. Click “Create subscription”.
157. Select the protocol.
158. Enter the endpoint, such as an email address.
159. Then add email
160. Create the subscription.

Step 4: Confirm Subscription

161. Open the confirmation message sent to the endpoint and confirm the subscription when required.

26. How to Create a CloudWatch Alarm

Step 1: Open CloudWatch

162. Search for “CloudWatch” and open it.

Step 2: Select Metric

163. Open “Alarms”.
164. Click “Create alarm”.
165. Select the required metric.

Step 3: Configure Condition

166. Set the threshold and evaluation settings.

Step 4: Configure Notification

167. Select an SNS topic for notifications if required.

Step 5: Create Alarm

168. Enter an alarm name.
169. Review the settings.
170. Click "Create alarm".

AWS CLI

1. How to Install AWS CLI

Step 1: Download AWS CLI

171. Download and install the AWS CLI for your operating system from the official AWS installation source.

Step 2: Verify Installation

172. Open a terminal or Command Prompt.
173. Run the AWS CLI version command to verify the installation.

28. How to Configure AWS CLI

Step 1: Open Terminal

174. Open Command Prompt, PowerShell, or a terminal.

Step 2: Configure AWS

175. Run the AWS CLI configure command.
176. Enter the AWS Access Key ID.
177. Enter the AWS Secret Access Key.
178. Enter the default AWS Region.
179. Enter the output format if required.

Step 3: Verify

180. Run an AWS CLI command to verify that the credentials and region are configured correctly.

Docker + ECR

Note : jo ma na rag ecommerce assistant wala project bana taa na aso ma hum ek docker ki file vanta ha jasi ma instruction hoti ha aur ci/cd file ma hum na code write kahi hoe ha docker build karna ka aur tag aur push karna ka waha sa check kar lana just 2 ya 3 lines ha

29. How to Create an ECR Repository

Step 1: Open ECR

181. Search for “ECR” in the AWS Console and open it.

Step 2: Create Repository

- 182. Click “Create repository”.
- 183. A new page will open.
- 184. Enter the repository name.
- 185. Don't change any other settings

Step 3: Create

186. Click “Create repository”.

ECS Cluster + Task Definition + Run Task

31. How to Create an ECS Cluster

Step 1: Open ECS

- 187. Search for “ECS” and open it.
- 188. Select “Clusters”.
- 189. Click “Create cluster”.

Step 2: Configure Cluster

- 190. Enter the cluster name.
- 191. Select the required infrastructure option, such as AWS Fargate, when appropriate.

Step 3: Create

- 192. Review the settings.
- 193. Click “Create”.

32. How to Create an ECS Task Definition

Step 1: Open Task Definitions

- 194. Open ECS → Task definitions.
- 195. Click “Create new task definition”.

Step 2: Configure Task

- 196. Enter the task family name.
- 197. Select the required launch type/infrastructure.
- 198. Configure CPU and memory.

Step 3: Configure Container

- 199. Add the container.

- 200. Enter the ECR image URI.
- 201. Configure the container port.
- 202. Add required environment variables and other settings.

Step 4: Create

- 203. Review the task definition.
- 204. Click "Create".

33. How to Run an ECS Task

Step 1: Open Cluster

- 205. Open ECS → Clusters.
- 206. Select the required cluster.

Step 2: Create Service/Run Task

- 207. Choose the option to create a service or run a task.
- 208. Select the task definition.
- 209. Configure the required networking and security group.

Step 3: Run

- 210. Configure the required subnet and public/private networking.
- 211. Create the service or run the task.

Step 4: Verify

- 212. Open the running task.
- 213. Check its status, logs, and networking information.

Docker → ECR → ECS

38. AWS Application Deployment Flow: Docker → ECR → ECS

Step 1: Prepare Application(project)

- 214. Create and test the application locally.

Step 2: Create Docker Image

- 215. Create a Dockerfile.
- 216. Build the Docker image.(this image every time we will make in ci/cd file)

Step 3: Create ECR Repository

- 217. Create an ECR repository for the application image.

Step 4: Push Image(also these steps in ci/cd file)

- 218. Authenticate Docker with ECR.
- 219. Tag the image with the ECR repository URI.

- 220. Push the image to ECR.

Step 5: Create ECS Resources

- 221. Create an ECS cluster.
- 222. Create a task definition using the ECR image.

Step 6: Deploy

- 223. Create an ECS service/task.
- 224. Configure networking and security groups.
- 225. Run the application.

Step 7: Verify

- 226. Check the ECS task/service status.
- 227. Check application logs.
- 228. Open the application endpoint when available.

39. AWS Resource Cleanup

Step 1: Stop/Terminate Compute Resources

- 229. Stop or terminate unused EC2 instances.
- 230. Remove unused ECS services/tasks.

Step 2: Delete Storage Resources

- 231. Delete unused S3 objects/buckets when they are no longer required.
- 232. Delete unused EBS volumes and snapshots when appropriate.

Step 3: Delete Databases

- 233. Delete unused RDS databases after confirming that the data is no longer required.

Step 4: Remove Networking Resources

- 234. Remove unused NAT Gateways, load balancers, Elastic IPs, and other resources when no longer required.

Step 5: Remove Infrastructure-as-Code Resources

- 235. If resources were created with Terraform, use terraform destroy when the complete environment is no longer required.